

Private Pilot Maneuver Guide

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Note: This guide is written to be used with any single-engine piston trainer. Wherever a speed, power setting, or configuration item is airplane-specific, reference your aircraft's checklist/POH rather than a fixed number.

1. Clearing Turns

Before every maneuver: clear the area with two 90° turns (or a 180°), check for traffic, terrain, and altitude — never skip this step.

2. Steep Turns

Purpose: Coordination and altitude/bank control at high bank angles.

1. Clear the area.
2. Set cruise power and maintain the recommended steep turn airspeed for your aircraft.
3. Choose and bug a heading/landmark.
4. Get trimmed out at the chosen altitude.
5. Roll into a 45° bank using proper rudder to keep the ball centered.
6. Through 30° of bank, add a touch of power to help hold altitude — roughly 150-200 RPM.
7. Apply nose-up trim as needed — roughly two full rolls of down (nose up) trim is typical.
8. Maintain altitude, coordination, and airspeed, keeping the nose on the horizon while anticipating rollout.
9. Start rollout 10-15° early to end on the bugged heading.
10. Apply forward pressure on rollout to maintain altitude.
11. Level out.
12. Repeat the maneuver in the opposite direction.

Tolerances: altitude ± 100 ft, airspeed ± 10 kts/MPH, bank $\pm 10^\circ$, rollout heading $\pm 10^\circ$.

Common errors: losing altitude in the turn, over-banking, skidding/slipping from uncoordinated rudder.

3. Entering Slow Flight

Purpose: Aircraft control at minimum controllable airspeed.

1. Clear the area.
2. Carb heat ON (if equipped/required).

3. Reduce power — roughly 1500-1700 RPM depending on aircraft — and configure flaps per your checklist as speed allows. Altitude can be maintained while doing this, with the nose pushed down as flaps are added. It's acceptable to add all flaps in at once, as long as you're within the flap operating range (V_{fe}).
4. Reference your checklist's power-off stall speed (V_{so}) and slow to approximately 5 kts/MPH above it.
5. Add power back in — roughly 2000 RPM is a good starting point. The best technique is to pitch for airspeed first, then add power to catch or maintain altitude. The power should be added back in until the altitude needle stabilizes while pitched for the desired airspeed.
6. Practice turns, climbs, and descents at this speed while holding altitude/heading.

Recovery:

1. Carb heat OFF (if applicable).
2. Full power.
3. Maintain forward pressure and altitude while slowly retracting flaps.

Tolerances: altitude ± 100 ft, heading $\pm 10^\circ$, airspeed +10/-0 kts/MPH (never let it decay below target).

Common errors: letting airspeed decay into a stall, uncoordinated flight, altitude bleed during turns.

4. Climbing in Slow Flight

Purpose: Practicing a climb while remaining at minimum controllable airspeed.

1. Carb heat — push in (OFF).
2. Apply full power.
3. Continue pitching for the same airspeed you've been maintaining.
4. If the airplane does not climb, a small amount of flaps can be taken out (and later re-added once established back in cruise).
5. Once the climb is complete, carb heat comes back out (ON).

Tolerances: heading $\pm 10^\circ$, airspeed +10/-0 kts/MPH.

Common errors: letting the airspeed decay while trying to force a climb, forgetting to remove carb heat before adding power.

5. Descending in Slow Flight

Purpose: Practicing a descent while remaining at minimum controllable airspeed.

1. Remove some power.
2. Continue pitching for the same airspeed.
3. The more power you take out, the faster you will descend.
4. If only a small descent is requested (100-200 ft), avoid taking out large amounts of power, as you may blow through your level-off at the desired altitude.

Tolerances: heading $\pm 10^\circ$, airspeed +10/-0 kts/MPH.

Common errors: removing too much power for a small descent and overshooting the level-off altitude, letting the airspeed decay.

6. Leveling Off from a Climb or Descent in Slow Flight

Purpose: Returning to level flight at the target altitude without ballooning through it or sinking below it.

1. As the target altitude approaches, promptly add or remove power to arrest the climb/descent — pitch stays roughly the same throughout, only power changes.
2. Lead the level-off slightly early, adjusting the timing based on how fast you were climbing or descending — the faster the rate, the earlier you'll need to make the adjustment.
3. Tip: return power to your baseline slow flight setting (roughly 2000 RPM) and fine-tune from there to hold the new altitude.

Tolerances: altitude ± 100 ft, airspeed +10/-0 kts/MPH.

Common errors: waiting too long to adjust power and blowing through the target altitude, changing pitch instead of power to level off.

7. Turning in Slow Flight

Purpose: Practicing turns while remaining at minimum controllable airspeed.

1. Turns should be done at a radius no greater than standard rate — the miniature airplane on the turn coordinator should be touching the second (bottom) line at most.
2. Power may need to be added in the turn to maintain altitude.
3. Continue pitching for the same airspeed you've been maintaining throughout the turn.

Tolerances: altitude ± 100 ft, airspeed +10/-0 kts/MPH, bank at or below standard rate (turn coordinator wing on or inside the second line).

Common errors: exceeding standard rate bank, letting airspeed or altitude decay in the turn.

8. Power-Off Stall (landing configuration)

Purpose: Simulates a stall on approach to landing.

1. Clear the area.
2. Bug a heading.
3. Get set up in slow flight configuration — for this setup, altitude does not need to be maintained, and the nose can be pushed down while flaps are added. It's acceptable to add all flaps in at once, as long as you're within the flap operating range (Vfe).
4. Initiate a descent at approximately 500 FPM.
5. Power to idle, bring the nose to the horizon.
6. Pull back more as the airplane slows down.

7. Airplane stalls.

Recovery:

1. Pitch nose down.
2. Full power.
3. Carb heat OFF (if applicable).
4. Retract 10° of flaps immediately.
5. Hold nose down and speed up to your checklist's Vx or Vy.
6. Climb at that speed.
7. When the vertical speed needle reverses, continue retracting flaps while climbing.

Tolerances: altitude ± 100 ft, heading $\pm 10^\circ$ (wings level entry) or bank $\pm 10^\circ$ (turning entry), recover promptly at the first indication of a stall (buffet, horn, or break).

9. Power-On Stall (departure/takeoff configuration)

Purpose: Simulates a stall during takeoff or climb.

1. Clear the area.
2. Bug a heading.
3. Reduce power to idle and maintain back pressure while slowing to your checklist's rotation speed.
4. At rotation speed, apply full power while pitching the nose up 25-30° on the attitude indicator.
5. Maintain coordination with rudder to prevent yaw and use ailerons to prevent roll.
6. Airplane stalls.

Recovery:

1. Pitch nose down to below the horizon.
2. Full power.
3. Hold until you achieve your checklist's Vx, then climb at that speed.

Tolerances: altitude ± 100 ft, heading $\pm 10^\circ$ (wings level entry) or bank $\pm 10^\circ$ (turning entry), recover promptly at the first indication of a stall.

Common errors: over-rotating too fast (secondary stall), picking up a dropping wing with aileron instead of rudder.

10. Engine-Out Procedure

Purpose: Practicing an emergency forced-landing sequence.

Flow method: floor to door — start at the fuel selector (floor), then work across the panel from the right side toward the left, ending at the door. Follow your aircraft's own checklist for the exact item order and switch positions; the flow below is the general sequence most checklists follow.

Engine Failure in Flight

1. Airspeed — best glide (Vg) per your checklist.
2. Carb heat — ON (if equipped).
3. Fuel selector valve — BOTH/appropriate tank.
4. Flaps — UP.
5. Mixture — RICH.
6. Primer — IN AND LOCKED.
7. Magnetos — CHECK ALL.

If prop not windmilling:

8. Ignition switch — START.
9. Magnetos — CHECK ALL.
10. Declare emergency.
11. Secure engine.

Engine Secure / Power Off Landing

1. Landing area — select & inspect.
2. Airspeed — best glide (Vg).
3. Mixture — CUTOFF.
4. Fuel selector valve — OFF.
5. Ignition & electrical switch — OFF.
6. Flaps — as required.
7. Master switch — OFF.
8. Doors — unlatch (prior to touchdown).
9. Brakes — apply heavily.

Tolerances: establish and hold best glide speed ± 10 kts/MPH throughout the glide, land within/near the selected touchdown area.

Common errors: fixating on the restart flow instead of flying the airplane and picking the landing site, forgetting to establish/hold best glide speed, and skipping through the panel out of order instead of running the floor-to-door flow.

11. S-Turns Across a Road

Purpose: Wind-drift correction, ground reference maneuver.

1. Listen to the weather (ATIS/AWOS or wind check) to get the wind direction.
2. Bug the wind direction, then identify a straight road/line running perpendicular to it. Position yourself with a tailwind so the heading bug sits on the bottom of the heading indicator.
3. Descend to the desired maneuvering altitude (typically 600-1000 ft AGL) — around Easterwood, this is roughly 1300 ft indicated.
4. Cross the road with wings level, roll immediately into a turn away from the wind.

5. Vary bank angle continuously — steepest bank at the downwind crossing point (fastest groundspeed), shallowest into the wind — to hold a constant radius/track shape on each side of the road. The first turn, entered with a tailwind, will be the steepest bank of the maneuver — this doesn't mean it's a "steep turn" in the 45°+ sense, just steeper than the turns that follow. The second turn (the shallow one) will accordingly be a longer half-S, or wider bowl shape, than the first.
6. Roll out wings-level exactly over the road each crossing, then roll into the opposite turn.
7. Maintain constant altitude throughout.

Tolerances: altitude ± 100 ft, airspeed ± 10 kts/MPH, symmetrical track on each side of the road, wings level exactly over the road at each crossing.

Common errors: crossing the road at an angle instead of perpendicular, constant bank angle (not adjusting for wind).

12. Turns Around a Point

Purpose: Wind-drift correction while maintaining a constant radius around a fixed ground point.

1. Listen to the weather (ATIS/AWOS or wind check) to get the wind direction.
2. Bug the wind direction, then turn to the heading 180° opposite so you're flying with a tailwind.
3. Descend to the desired maneuvering altitude (typically 600-1000 ft AGL).
4. Flying downwind, find a prominent ground point that falls naturally off the side of the wing to use as your pivot point.
5. Establish a bank (steepest abeam downwind, shallowest abeam upwind) to maintain a constant radius from the point through all 360°. Since you enter with a tailwind, the initial bank will be the steepest of the maneuver — this doesn't mean it's a "steep turn" in the 45°+ sense, just steeper than the bank used through the rest of the circle.
6. Continuously adjust bank as groundspeed changes around the circle.
7. Maintain constant altitude.
8. Roll out on the original entry heading after one or more full circles.

Tolerances: altitude ± 100 ft, constant radius from the point, rollout heading $\pm 10^\circ$.

Common errors: radius drifting in/out (not adjusting bank quickly enough), altitude loss/gain, not correcting for changing wind correction angle throughout the turn.

General Notes

- All maneuvers: clear the area first, and maintain coordinated flight (ball centered) unless intentionally practicing slips.

- Carb heat: use per your aircraft's checklist anytime power is reduced significantly (slow flight, stalls, engine-out practice) on carbureted engines.
- These are training-reference outlines, not a substitute for your POH, instructor guidance, or the ACS/PTS standards you're being evaluated against.